Linear Acoustics with Advection and Multirate Time Integration

Split-explicit, exponential, Lawson, averaging and resonance-based methods

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October 20, 2026

Outline

1. Motivation: time integration in NWP
2. Model problem
3. Split-explicit methods as ETD
4. Lawson, averaging and resonance methods
5. Application: staggered grids, high-order advection
6. Summary

Why this problem? Time stepping in numerical weather models

- Compressible (non-hydrostatic) dynamical cores solve the Euler equations on the sphere/in a box.
- Wide range of time scales in one system:

process	speed	role
sound waves	$c \approx 340\,\mathrm{m/s}$	fast, *not* meteorologically relevant
gravity waves	~ 30–$100\,\mathrm{m/s}$	fast/intermediate
advection by wind	$U \approx 10$–$50\,\mathrm{m/s}$	slow, *relevant*

- Fully explicit methods: step size limited by $c\,\Delta t/\Delta x \lesssim 1 \;\Rightarrow$ far smaller than accuracy needs for the slow, relevant dynamics.
- Fully implicit: global (Helmholtz-type) solves, poor scalability on massively parallel machines.

Key idea

Treat the fast part cheaply/exactly and the slow part with a large step: **multirate** / **split** / **exponential** methods.

Where it is used and what is needed

- Split-explicit time stepping (Klemp–Wilhelmson; Wicker–Skamarock RK3 with acoustic substeps) is the standard in WRF, COSMO/ICON-type and many other cores.
- Exponential integrators and multirate schemes are active research directions for dynamical cores.
- The **linear acoustic–advection system** is the core subproblem:
  - it contains exactly the scale separation $c \gg U$ (Mach number $M = U/c \approx 0.1$),
  - it allows analysis in Fourier space (stability, dispersion, resonance),
  - errors of any scheme for the full nonlinear system must at least be good here.

Plan of the talk

1. model problem and the abstract splitting $\mathrm{q}' = \mathcal{L}\mathrm{q} + \mathcal{N}\mathrm{q}$
2. split-explicit methods as a special case of ETD
3. Lawson, averaging and resonance-based methods
4. application: staggered grids in 1D and 2D with high-order advection

Linear acoustics with advection

Perturbations (p, u) about a constant background state (ρ_0, U), speed of sound c:

$$\partial_t u + U\,\partial_x u + \tfrac{1}{\rho_0}\,\partial_x p = 0,$$
$$\partial_t p + U\,\partial_x p + \rho_0 c^2\,\partial_x u = 0.$$

Write $\mathrm{q} = (p, u)^\top$ and split

$$\partial_t \mathrm{q} = \underbrace{-\begin{pmatrix} 0 & \rho_0 c^2 \\ 1/\rho_0 & 0 \end{pmatrix}\partial_x \mathrm{q}}_{\mathcal{L}\mathrm{q}:\ \text{acoustic (fast)}} \underbrace{-\,U\,\partial_x \mathrm{q}}_{\mathcal{N}\mathrm{q}:\ \text{advection (slow)}}.$$

- $\mathcal{L}$: eigenvalues in Fourier space $\pm \mathrm{i}ck$ (purely oscillatory, large).
- $\mathcal{N}$: eigenvalue $-\mathrm{i}Uk$ (small, $|U| \ll c$).
- For constant U on a periodic domain the two operators **commute**, also after discretization with the same stencil for p and u (all operators are circulant). Non-commutativity enters via variable $U(x)$, boundaries, or flux-form/interpolated discretizations.

Abstract setting

$$\mathrm{q}'(t) = \mathcal{L}\mathrm{q}(t) + \mathcal{N}\mathrm{q}(t), \qquad \mathrm{q}(t_0) = \mathrm{q}_0,$$

after spatial discretization: $\mathcal{L}, \mathcal{N}$ are matrices, $\|h\mathcal{L}\| \gg \|h\mathcal{N}\|$ for the time steps of interest.

Variation of constants over one step $[t_n, t_n + h]$:

$$\mathrm{q}(t_n + h) = \mathrm{e}^{h\mathcal{L}}\mathrm{q}_n + \int_0^h \mathrm{e}^{(h-s)\mathcal{L}}\,\mathcal{N}\,\mathrm{q}(t_n + s)\,\mathrm{d}s.$$

- The fast part is treated *exactly* (or very accurately) through $\mathrm{e}^{h\mathcal{L}}$.
- Only the slow part $\mathcal{N}\mathrm{q}$ must be approximated; its time scale is $\sim 1/(Uk)$.
- Different multirate methods = different ways to approximate this integral.

Notation: $\varphi_1(z) = \dfrac{\mathrm{e}^z - 1}{z}, \quad \varphi_2(z) = \dfrac{\varphi_1(z) - 1}{z}, \quad \varphi_{k+1}(z) = \dfrac{\varphi_k(z) - 1/k!}{z}.$

Exponential time differencing (ETD)

Approximate $\mathcal{N}\mathsf{q}(t_n + s)$ by a polynomial in s and integrate the exponential exactly.

ETD1 (exponential Euler):

$$\mathsf{q}_{n+1} = \mathrm{e}^{h\mathcal{L}}\mathsf{q}_n + h\,\varphi_1(h\mathcal{L})\,\mathcal{N}\mathsf{q}_n.$$

ETD2 / ETDRK2 (Cox–Matthews type):

$$\begin{aligned}\mathsf{a}_n &= \mathrm{e}^{h\mathcal{L}}\mathsf{q}_n + h\,\varphi_1(h\mathcal{L})\mathcal{N}\mathsf{q}_n,\\ \mathsf{q}_{n+1} &= \mathsf{a}_n + h\,\varphi_2(h\mathcal{L})\big(\mathcal{N}\mathsf{a}_n - \mathcal{N}\mathsf{q}_n\big).\end{aligned}$$

- Exact for $\mathsf{q}' = \mathcal{L}\mathsf{q} + \mathsf{g}$ with constant g ("frozen slow tendency").
- Requires $\mathrm{e}^{h\mathcal{L}}\mathsf{v}$ and $\varphi_k(h\mathcal{L})\mathsf{v}$: Krylov / Fourier / (rational) approximations.
- Acoustic oscillations: no CFL restriction from c at all (but see resonances later).

Split-explicit time stepping

Klemp–Wilhelmson / Wicker–Skamarock: freeze the slow tendency during the step and integrate the fast system by m small substeps $\tau = h/m$ (forward–backward for the acoustic part):

$$\mathsf{z}_0 = \mathsf{q}_n, \qquad \mathsf{z}_{j+1} = \mathsf{z}_j + \tau\big(\mathcal{L}\mathsf{z}_j + \mathsf{g}\big), \quad \mathsf{g} = \mathcal{N}\mathsf{q}_n, \qquad \mathsf{q}_{n+1} = \mathsf{z}_m.$$

Since $\mathsf{z}' = \mathcal{L}\mathsf{z} + \mathsf{g}$ has the exact solution $\mathrm{e}^{t\mathcal{L}}\mathsf{q}_n + t\,\varphi_1(t\mathcal{L})\mathsf{g}$,

$$\begin{aligned} \mathsf{z}_m &= (I + \tau\mathcal{L})^m \mathsf{q}_n + \tau \sum_{j=0}^{m-1} (I + \tau\mathcal{L})^j \,\mathsf{g} \\ &\approx \mathrm{e}^{h\mathcal{L}}\mathsf{q}_n + h\,\varphi_1(h\mathcal{L})\mathsf{g}. \end{aligned}$$

Split-explicit = ETD1 with substepped approximations

$\mathrm{e}^{h\mathcal{L}} \;\to\; R_m(\tau\mathcal{L}) = (I + \tau\mathcal{L})^m$ and $\varphi_1(h\mathcal{L}) \;\to\; \frac{1}{m}\sum_{j<m}(I + \tau\mathcal{L})^j$. The same view applies to the RK3 outer loop of Wicker–Skamarock (stages with $h/3$, $h/2$, h).

Wicker–Skamarock RK3 (WRF) as an exponential method

$S_t(\mathrm{q};\mathrm{g})$: forward–backward substeps over $[0,t]$ for $\mathrm{z}' = \mathcal{L}\mathrm{z} + \mathrm{g}$, $\mathrm{z}(0) = \mathrm{q}$ $\qquad (\approx \mathrm{e}^{t\mathcal{L}}\mathrm{q} + t\,\varphi_1(t\mathcal{L})\mathrm{g})$

$$\mathrm{q}^* = S_{h/3}(\mathrm{q}_n; \mathcal{N}\mathrm{q}_n), \qquad \mathrm{q}^{**} = S_{h/2}(\mathrm{q}_n; \mathcal{N}\mathrm{q}^*), \qquad \mathrm{q}_{n+1} = S_h(\mathrm{q}_n; \mathcal{N}\mathrm{q}^{**}).$$

WRF: n_s steps $\Delta\tau = h/n_s$ per h; stage 1 takes *one* step $h/3$, stage 2 takes $n_s/2$ steps. **Forward–backward with pressure extrapolation** (Klemp–Skamarock–Dudhia 2007):

$$\tilde{p} = p^\tau + \beta\,(p^\tau - p^{\tau-\Delta\tau}), \qquad u^{\tau+\Delta\tau} = u^\tau - \Delta\tau\big(\tfrac{1}{\rho_0} D\tilde{p} - g_u\big), \qquad p^{\tau+\Delta\tau} = p^\tau - \Delta\tau\big(\rho_0 c^2 D u^{\tau+\Delta\tau} - g_p\big).$$

Since $p^\tau - p^{\tau-\Delta\tau} \approx -\Delta\tau\rho_0 c^2 Du$, this is divergence damping with coefficient $\beta c^2 \Delta\tau$ (WRF: $\beta = 0.1$).

Coupling order (exact fast part, one Fourier mode $\mathcal{L} \to \mathrm{i}\omega$, $\mathcal{N} \to \nu$, $\theta = \omega h$):

$$R = \mathrm{e}^{\mathrm{i}\theta} + h\nu\,\varphi_1(\mathrm{i}\theta)\,\mathrm{e}^{\mathrm{i}\theta/2} + \mathcal{O}(h^2\nu^2), \qquad \varphi_1(\mathrm{i}\theta)\,\mathrm{e}^{\mathrm{i}\theta/2} = \mathrm{e}^{\mathrm{i}\theta}\,\frac{\sin(\theta/2)}{\theta/2}.$$

- Local error $h\nu\,\mathrm{e}^{\mathrm{i}\theta}\big(\frac{\sin(\theta/2)}{\theta/2} - 1\big) \approx -\frac{1}{24} h\nu\theta^2$: the frozen tendency makes the scheme **second order** in h (the 3rd order MIS coupling conditions fail), and the error is $\mathcal{O}(1)$ for $\theta \sim \pi$.
- Lawson-RK4 and ETDRK4 are exact in this term.

Multirate infinitesimal step (MIS) methods

Generalized split-explicit RK (Wensch–Knoth–Galant 2009, Knoth–Wensch 2014): stages start from and are forced by combinations of *all* previous stages,

$$\mathsf{Z}_i(0) = \mathsf{q}_n + \sum_{j<i} \alpha_{ij}\,(\mathsf{Y}_j - \mathsf{q}_n), \qquad \mathsf{Z}_i'(t) = \frac{1}{h}\sum_{j<i} \gamma_{ij}\,(\mathsf{Y}_j - \mathsf{q}_n) + \sum_{j<i} \beta_{ij}\,\mathcal{N}\mathsf{Y}_j + d_i\,\mathcal{L}\mathsf{Z}_i(t), \quad t \in [0,h],$$

$$\mathsf{Y}_i = \mathsf{Z}_i(h), \qquad d_i = \sum_j \beta_{ij}, \qquad \mathsf{q}_{n+1} = \mathsf{Y}_{s+1}.$$

ETD form: $\mathsf{Y}_i = \mathrm{e}^{d_i h\mathcal{L}}\mathsf{Z}_i(0) + h\,\varphi_1(d_i h\mathcal{L})\,\mathsf{F}_i$: **one pass of the fast integrator** with constant source F_i/d_i over $d_i h$. For $\mathcal{L} = 0$: the RK method $A = (I - \alpha - \Gamma)^{-1}\beta$.

method		stages	fast time per step $\sum_i d_i\,h$	order
RK3 (Wicker–Skamarock)	$\alpha = \Gamma = 0$	3	$1.83\,h$	2
MIS-KW3 (Knoth–Wolke RK3)	$\alpha_{i,i-1} = 1$, $\Gamma = 0$	3	$1.00\,h$	3
MIS54 (Bauer–Knoth 2021)	full α, β, Γ	5	$1.88\,h$	4

- MIS-KW: $\mathsf{Z}_i(0) = \mathsf{Y}_{i-1}$, i.e. recursive stages over $[c_{i-1}h, c_i h]$; KW3: $c = (0, \frac{1}{3}, \frac{3}{4})$, $b = (\frac{1}{6}, \frac{3}{10}, \frac{8}{15})$.
- Order 3 and 4 need extra coupling conditions; the fast stages may be integrated by any one-step method of the same order (Bauer–Knoth 2021).

Consequences of the ETD view

- **Order:** the slow forcing is frozen $\Rightarrow$ first order for the pure ETD1 version; higher order through RK-type outer stages or ETDRK schemes with φ_2, φ_3.
- **Fast part inexact:** $\mathrm{e}^{h\mathcal{L}}$ is replaced by a stability-limited polynomial; the substep count m controls both accuracy and stability (acoustic Courant number $c\tau/\Delta x \leq C$).
- **Free choice of "fast solver"**: substepping, Fourier/FFT (periodic), Krylov, Chebyshev – all realize the same ETD framework. Here: *one* substepping routine delivers every $\mathrm{e}^{\cdot\mathcal{L}}$ and φ_k.
- **High order needs a high-order fast part**: for large slow steps h the acoustic substep τ is fixed by stability, so the fast error does not shrink with h (next two slides, and the experiments).
- **Analysis in Fourier space:** on each mode, $\mathcal{L} \to \pm \mathrm{i}ck$, $\mathcal{N} \to -\mathrm{i}Uk$ and all $\varphi_k(\mathrm{i}\theta)$ with $\theta = \pm ckh$ are explicit; amplification factors can be compared with $\mathrm{e}^{-\mathrm{i}(\pm c - U)\, kh}$.

Fast integrator: forward–backward kicks, $\mathrm{e}^{h\mathcal{L}}$ and φ_k in one pass

Fast part $\mathsf{z}' = \mathcal{L}\mathsf{z} + \mathsf{s}(t)$, $\mathcal{L} = \begin{pmatrix} 0 & -\rho c^2 D_x \\ -D_x/\rho & 0 \end{pmatrix}$, split into two exactly solvable *kicks*:

$$
\begin{aligned}
A(\alpha):&\quad p \leftarrow p - \alpha\,\rho c^2 D_x u + \textstyle\int_t^{t+\alpha} s_p\,\mathrm{d}s, \quad t \leftarrow t+\alpha && (u \text{ frozen})\\
B(\beta):&\quad u \leftarrow u - \beta\, D_x p/\rho + \beta\, s_u(t) && (p,\ t \text{ frozen})
\end{aligned}
$$

- forward–backward (Klemp–Wilhelmson): $B(\tau)A(\tau)$, first order; leapfrog $A(\frac{\tau}{2})B(\tau)A(\frac{\tau}{2})$, second order.
- **Polynomial source** $\mathsf{s}(t) = \sum_{k\ge1} \mathsf{G}_k\, t^{k-1}/(k-1)!$ gives

$$\mathsf{z}(h) = \mathrm{e}^{h\mathcal{L}}\mathsf{z}(0) + \sum_{k\ge1} h^k\, \varphi_k(h\mathcal{L})\,\mathsf{G}_k,$$

  so every exponential and every φ-function in the schemes comes from the *same* routine.
- Split-explicit step (ETD1) = one pass with $\mathsf{z}(0) = \mathsf{q}_n$, $\mathsf{G}_1 = \mathcal{N}\mathsf{q}_n$.
- Cost per substep: one A and one B kick (4th order staggered stencils).

High-order fast integrator: symmetric composition

Why: the acoustic substep is limited by stability, $c\,\tau\,\max_i |w_i|/\Delta x \lesssim 6/7$ (4th order stencil), and does *not* shrink when the slow step h is refined. A 2nd order fast part therefore caps the order of the whole scheme.

Strang step $\Psi_\tau = A(\frac{\tau}{2})B(\tau)A(\frac{\tau}{2})$ (symmetric, order 2). Composition with weights w_i, $\sum_i w_i = 1$: $\Psi_{w_1\tau}\Psi_{w_2\tau}\cdots\Psi_{w_s\tau}$ has order 4 ($s = 3$) or 6 ($s = 7$).

order	weights w_i	kicks per substep
2	1	$1A + 1B$ (leapfrog)
4	$\gamma_1, \gamma_2, \gamma_1$ with $\gamma_1 = \frac{1}{2 - 2^{1/3}} \approx 1.351$, $\gamma_2 = 1 - 2\gamma_1 \approx -1.702$	$3A + 3B$
6	$w_3, w_2, w_1, w_0, w_1, w_2, w_3$ (Yoshida): $w_1 \approx -1.178$, $w_2 \approx 0.236$, $w_3 \approx 0.785$, $w_0 = 1 - 2(w_1 + w_2 + w_3) \approx 1.315$	$7A + 7B$

- Neighbouring A-kicks (end of one Ψ, start of the next) are merged.
- Time-dependent sources are no problem: t advances in the A-kicks, the integral of the polynomial is exact.

Accuracy of the fast integrator ($h = 50\,\mathrm{s}$, $ch/\Delta x = 7$)

Error against the exact $\mathrm{e}^{h\mathcal{L}}$ and $\varphi_2(h\mathcal{L})$ (Fourier reference, same spatial operator); each column halves $\tau = h/\mathrm{nsub}$:

	order	nsub=16	32	64	128
$\mathrm{e}^{h\mathcal{L}}$	2	$1.1 \cdot 10^{-5}$	$2.6 \cdot 10^{-6}$	$6.6 \cdot 10^{-7}$	$1.6 \cdot 10^{-7}$
	4	$4.8 \cdot 10^{-8}$	$3.0 \cdot 10^{-9}$	$1.9 \cdot 10^{-10}$	$1.2 \cdot 10^{-11}$
	6	$4.1 \cdot 10^{-10}$	$6.3 \cdot 10^{-12}$	$9.8 \cdot 10^{-14}$	$2.2 \cdot 10^{-15}$
$\varphi_2(h\mathcal{L})$ (relative)	2	$2.0 \cdot 10^{-3}$	$4.9 \cdot 10^{-4}$	$1.2 \cdot 10^{-4}$	$3.1 \cdot 10^{-5}$
	4	$3.1 \cdot 10^{-8}$	$1.9 \cdot 10^{-9}$	$1.2 \cdot 10^{-10}$	$7.5 \cdot 10^{-12}$
	6	$8.6 \cdot 10^{-12}$	$1.3 \cdot 10^{-13}$	$3.0 \cdot 10^{-14}$	$3.7 \cdot 10^{-14}$

- Error factors per halving of τ: 4, 16, 64 for orders 2, 4, 6. Same for φ_1, φ_3.
- The 2nd order φ_2 error is large: a low-order fast part spoils the φ_k terms of ETD schemes first.

Table computed by `test_fast.jl`; the NumPy version of the algorithm gives the same numbers.

Lawson (integrating factor) methods

Change variables $\mathsf{v}(t) = \mathrm{e}^{-(t-t_n)\mathcal{L}}\mathsf{q}(t)$:

$$\mathsf{v}'(t) = \mathrm{e}^{-(t-t_n)\mathcal{L}}\,\mathcal{N}\,\mathrm{e}^{(t-t_n)\mathcal{L}}\mathsf{v}(t) =: \widetilde{\mathcal{N}}(t)\,\mathsf{v}(t),$$

apply an explicit Runge–Kutta method to v and transform back.

Lawson–Euler:

$$\mathsf{q}_{n+1} = \mathrm{e}^{h\mathcal{L}}\big(\mathsf{q}_n + h\mathcal{N}\mathsf{q}_n\big).$$

Lawson–RK4: classical RK4 with $\mathrm{e}^{c_i h\mathcal{L}}$ weights, e.g. stage values $\mathsf{Y}_i = \mathrm{e}^{c_i h\mathcal{L}}\mathsf{q}_n + h\sum_j a_{ij}\,\mathrm{e}^{(c_i-c_j)h\mathcal{L}}\mathcal{N}\mathsf{Y}_j$.

- Simple to implement: only matrix exponentials of $\mathcal{L}$ (diagonal in Fourier space).
- Order conditions are those of the underlying RK method (for sufficiently smooth solutions).
- Drawbacks: does not preserve steady states of $\mathcal{L}+\mathcal{N}$; error constants depend on $\|\mathcal{N}\|$, not on a well-adapted φ-function structure; stiff order reduction possible.

Efficient Lawson-RK4

Classical RK4 on $\mathrm{v}' = \widetilde{\mathcal{N}}(t)\mathrm{v}$ with $E = \mathrm{e}^{h\mathcal{L}/2}$ gives

$$\mathrm{q}_{n+1} = E^2\mathrm{q}_n + \tfrac{h}{6}\big(E^2\mathrm{k}_1 + 2E(\mathrm{k}_2 + \mathrm{k}_3) + \mathrm{k}_4\big),$$

with stages $\mathrm{Y}_1 = \mathrm{q}_n$, $\mathrm{Y}_2 = E(\mathrm{q}_n + \tfrac{h}{2}\mathrm{k}_1)$, $\mathrm{Y}_3 = E\mathrm{q}_n + \tfrac{h}{2}\mathrm{k}_2$, $\mathrm{Y}_4 = E^2\mathrm{q}_n + hE\mathrm{k}_3$, $\quad \mathrm{k}_i = \mathcal{N}\mathrm{Y}_i$.

Reorganized with E applied to sums (using linearity of E):

$$\begin{aligned}
&\mathrm{k}_1 = \mathcal{N}\mathrm{q}_n, && \mathrm{w} = E\mathrm{q}_n, && \mathrm{Y}_2 = E\big(\mathrm{q}_n + \tfrac{h}{2}\mathrm{k}_1\big), && \mathrm{k}_2 = \mathcal{N}\mathrm{Y}_2,\\
&\mathrm{Y}_3 = \mathrm{w} + \tfrac{h}{2}\mathrm{k}_2, && \mathrm{k}_3 = \mathcal{N}\mathrm{Y}_3, && \mathrm{Y}_4 = E\big(\mathrm{w} + h\mathrm{k}_3\big), && \mathrm{k}_4 = \mathcal{N}\mathrm{Y}_4,
\end{aligned}$$

$$\mathrm{q}_{n+1} = E\Big[\mathrm{w} + \tfrac{1}{3}\big(\mathrm{Y}_2 - \mathrm{w}\big) + \tfrac{h}{3}(\mathrm{k}_2 + \mathrm{k}_3)\Big] + \tfrac{h}{6}\mathrm{k}_4.$$

- **4 evaluations of $\mathcal{N}$ and 4 applications of E** per step; $\mathrm{e}^{h\mathcal{L}} = E^2$ is never formed ($\tfrac{1}{3}(\mathrm{Y}_2 - \mathrm{w}) = \tfrac{h}{6}E\mathrm{k}_1$ reuses Y_2).
- $E = \mathrm{e}^{h\mathcal{L}/2}$ is applied by the fast integrator over $h/2$ with nsub substeps. Only linearity of E is used, so the reorganization holds for the numerical E as well.

ETDRK4 with the same fast integrator

Cox–Matthews ETDRK4 with $g_i = \mathcal{N}(\text{stage } i)$; each stage is one pass of the fast integrator over $h/2$ with source G_1:

$$a = e^{\frac{h}{2}\mathcal{L}} q_n + \tfrac{h}{2}\varphi_1(\tfrac{h}{2}\mathcal{L})\, g_1, \qquad b = e^{\frac{h}{2}\mathcal{L}} q_n + \tfrac{h}{2}\varphi_1(\tfrac{h}{2}\mathcal{L})\, g_2,$$

$$c = e^{\frac{h}{2}\mathcal{L}} a + \tfrac{h}{2}\varphi_1(\tfrac{h}{2}\mathcal{L})\,(2g_3 - g_1), \qquad g_2 = \mathcal{N}a, \quad g_3 = \mathcal{N}b, \quad g_4 = \mathcal{N}c.$$

Final stage $q_{n+1} = e^{h\mathcal{L}} q_n + h\big[(\varphi_1 - 3\varphi_2 + 4\varphi_3)g_1 + 2(\varphi_2 - 2\varphi_3)(g_2 + g_3) + (-\varphi_2 + 4\varphi_3)g_4\big]$, regrouped by φ_k it is *one* pass over h with the polynomial source G_1, G_2, G_3:

$$G_1 = g_1, \qquad G_2 = \frac{-3g_1 + 2(g_2 + g_3) - g_4}{h}, \qquad G_3 = \frac{4(g_1 - g_2 - g_3 + g_4)}{h^2}.$$

- 4 evaluations of $\mathcal{N}$; fast integration: $3 \times \frac{h}{2} + h = 2.5\,h$ (Lawson-RK4: $4 \times \frac{h}{2} = 2\,h$).
- No φ-function is ever evaluated explicitly (no FFT, no matrix functions).

Averaging methods

Observation: in the interaction picture the slow forcing is

$$\widetilde{\mathcal{N}}(s) = \mathrm{e}^{-s\mathcal{L}}\mathcal{N}\,\mathrm{e}^{s\mathcal{L}},$$

which oscillates with the fast (acoustic) frequencies $\pm ck$. Its time average

$$\overline{\mathcal{N}}_h = \frac{1}{h}\int_0^h \mathrm{e}^{-s\mathcal{L}}\mathcal{N}\,\mathrm{e}^{s\mathcal{L}}\,\mathrm{d}s$$

carries the net slow effect; oscillating contributions average out.

Averaged integrator (first order sketch):

$$\mathsf{q}_{n+1} = \mathrm{e}^{h\mathcal{L}}\Big(I + h\overline{\mathcal{N}}_h\Big)\mathsf{q}_n \quad \text{or} \quad \mathsf{q}_{n+1} = \mathrm{e}^{h(\mathcal{L}+\overline{\mathcal{N}}_h)}\mathsf{q}_n.$$

- For commuting $\mathcal{L}, \mathcal{N}$: $\overline{\mathcal{N}}_h = \mathcal{N}$, and the scheme is exact in the continuum.
- For non-commuting operators (variable wind, interpolated fluxes, boundaries), $\overline{\mathcal{N}}_h$ captures the leading commutator corrections $[\mathcal{L}, \mathcal{N}]$.
- The average can be computed by quadrature in s, by a Fourier/eigenbasis formula, or by substeps.

Resonances

On a Fourier mode with $\mathcal{L} \to \pm \mathrm{i}\omega$, $\omega = ck$, the exponential integrator terms contain

$$\varphi_1(\mathrm{i}\theta) = \frac{\mathrm{e}^{\mathrm{i}\theta} - 1}{\mathrm{i}\theta}, \qquad \theta = \omega h,$$

and error constants contain factors like $1/\sin(\theta/2)$ in multistep/mean-value type schemes.

Step-size resonances

When $\omega h \approx 2\pi n$ (the step is a multiple of the acoustic period), errors can be amplified (in the worst case, instability) for some exponential/Lawson schemes.

Resonance-based (filtered) methods:

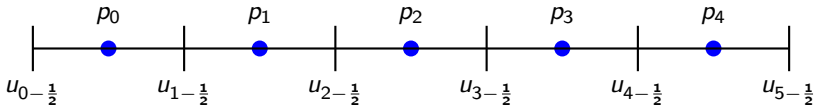
- expand the Duhamel integral and integrate the oscillatory factors $\mathrm{e}^{\mathrm{i}s(\pm\omega_1 \pm \omega_2)}$ *exactly*;
- resonant combinations ($\omega_1 \pm \omega_2 \approx 0$) are kept as they are, non-resonant ones are integrated analytically;
- result: error bounds that do not depend on ωh and no step-size resonance.

Comparison of the schemes (what we will test)

	fast part	slow part	resonance
split-explicit	FB substeps (1 pass)	frozen / RK stages	none (stability limit)
ETD(RK)	$\mathrm{e}^{\cdot\mathcal{L}}$, φ_k by fast integrator	polynomial in s	possible
Lawson	$\mathrm{e}^{h\mathcal{L}/2}$ by fast integrator	RK on interaction picture	possible
averaging	$\mathrm{e}^{h\mathcal{L}}$ by fast integrator	time-averaged $\overline{\mathcal{N}}_h$	reduced
resonance-based	$\mathrm{e}^{h\mathcal{L}}$ by fast integrator	oscillatory integrals exact	avoided

- Criteria: order in h, accuracy vs. ωh (large acoustic steps), cost (number of $\mathcal{L}$-evaluations / FFTs / substeps), robustness near $ckh = 2\pi n$.

Spatial discretization: staggered grid (1D)



- p at cell centers, u at cell faces (Arakawa C in 2D): compact, accurate acoustic operator

$$(\partial_x u)_j \approx \frac{u_{j+1/2} - u_{j-1/2}}{\Delta x}, \qquad (\partial_x p)_{j+1/2} \approx \frac{p_{j+1} - p_j}{\Delta x} \quad \text{(2nd order)}.$$

- Higher order: wider centered staggered stencils (4th/6th order) for $\mathcal{L}$.
- **Advection** $\mathcal{N} = -U\partial_x$: both fields must be advected on their own grid points $\Rightarrow$ high-order upwind-biased stencils (e.g. 3rd/5th order). With constant U and the same stencil on both arrays, $\mathcal{L}$ and $\mathcal{N}$ are circulant and commute: the exact semi-discrete solution is known (used as reference below).

High-order advection operators

Upwind-biased reconstruction of face values $\hat{\phi}_{j+1/2}$ (for $U > 0$), advective operator

$$(\mathcal{N}\phi)_j = -\frac{U}{\Delta x}\left(\hat{\phi}_{j+1/2} - \hat{\phi}_{j-1/2}\right).$$

order	face reconstruction
1st	$\hat{\phi}_{j+1/2} = \phi_j$
3rd	$\hat{\phi}_{j+1/2} = \frac{1}{6}\left(-\phi_{j-1} + 5\phi_j + 2\phi_{j+1}\right)$
5th	$\hat{\phi}_{j+1/2} = \frac{1}{60}\left(2\phi_{j-2} - 13\phi_{j-1} + 47\phi_j + 27\phi_{j+1} - 3\phi_{j+2}\right)$

- Odd orders: built-in dissipation (upwinding) $\Rightarrow$ implicit damping of grid-scale noise.
- Time integrator error must stay below the spatial error: low-order splitting would spoil the high-order advection.
- Fourier symbol of the discrete operator: $\mathcal{N} \to -\mathrm{i}U\,\kappa(k\Delta x)/\Delta x$, used for analysis of all schemes above.

Test setup: 1D

- Periodic domain $[0, L]$, background U, c with $M = U/c \in \{0.05, 0.1, 0.3\}$.
- Smooth initial data: Gaussian pressure pulse (right/left moving acoustic waves + advection).
- Reference: exact semi-discrete solution (circulant operators commute: analytic in Fourier space).
- $L = 1000\,\mathrm{km}$, 400 cells, $c = 350\,\mathrm{m/s}$, $U = 50\,\mathrm{m/s}$; slow steps $h = 50 \to 3\,\mathrm{s}$ ($ch/\Delta x = 7 \to 0.4$).
- Study (all exponentials and φ-functions from the fast integrator):
  1. pressure pulse: out and back (Lawson-RK4, fast order 4),
  2. convergence in h with the acoustic substep τ fixed (stability limited): fast order 2 vs. 4, Lawson-RK4 vs. ETDRK4, and the split-explicit RK3 of WRF (forward–backward, with and without pressure extrapolation),
  3. MIS methods (RK3, MIS-KW3, MIS54) and the linear acoustics stability test of Knoth–Wensch,
  4. work–precision: fast order 2, 4, 6 at large and at small slow steps.

Julia: the fast integrator (`fast_fb.jl`, abbreviated)

```
function kickA!(p, u, m, alpha, t, Gp) # exact p-kick, time advances by alpha
    n = m.nx; a = -alpha*m.rho*m.c^2/m.dx
    for j in 1:n
        d = 9/8*(u[wrap(j+1,n)] - u[j]) - 1/24*(u[wrap(j+2,n)] - u[wrap(j-1,n)])
        p[j] += a*d
    end
    for k in 1:length(Gp) # polynomial source: exact time integral
        w = ((t+alpha)^k - t^k)/factorial(k); gk = Gp[k]; @. p += w*gk
    end
end # kickB! (u-kick, t frozen) is analogous

function fast!(p, u, m, h, nsub, cf, Gp, Gu) # (p,u) <- e^{hL}(p,u) + sum_k h^k phi_k(hL) G_k
    tau = h/nsub; n = length(cf.b); t = 0.0
    pend = cf.a[1]*tau
    for s in 1:nsub
        if s > 1; pend += cf.a[1]*tau; end # merge neighbouring A-kicks
        for i in 1:n
            kickA!(p, u, m, pend, t, Gp); t += pend
            kickB!(u, p, m, cf.b[i]*tau, t, Gu)
            pend = cf.a[i+1]*tau
        end
    end
    kickA!(p, u, m, pend, t, Gp)
end
```

- `cf = Coeffs(order)` with `order` $\in \{2, 4, 6\}$; `Gp`, `Gu`: vectors of arrays $G_1, G_2, \ldots$ (empty for a pure exponential).

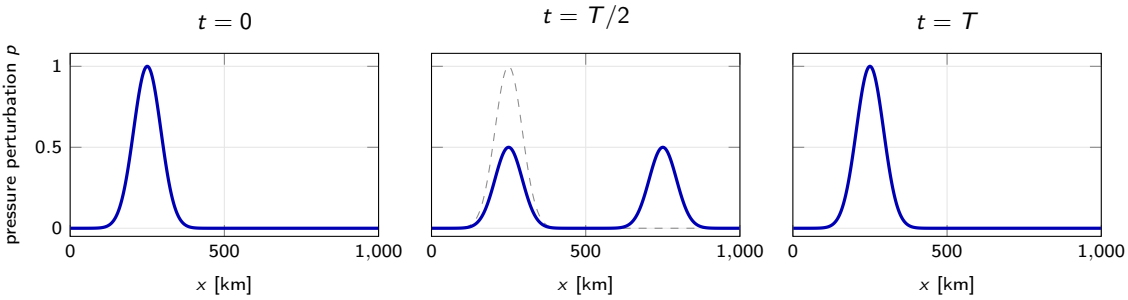
Julia: Lawson-RK4 step (4 evaluations of $\mathcal{N}$, 4 applications of E)

```julia
function lawson_rk4_step!(p, u, m, h, nsub, cf, ws)
    fh = ws.fh
    E!(a, b) = fast_exp!(a, b, m, h/2, nsub, cf) # E = exp(hL/2)
    adv!(ws.ap, p, m, fh); adv!(ws.au, u, m, fh) # k1
    copyto!(ws.wp, p); copyto!(ws.wu, u); E!(ws.wp, ws.wu) # w = E y
    @. ws.y2p = p + h/2 * ws.ap
    @. ws.y2u = u + h/2 * ws.au
    E!(ws.y2p, ws.y2u) # Y2 = E(y + h/2 k1)
    adv!(ws.bp, ws.y2p, m, fh); adv!(ws.bu, ws.y2u, m, fh) # k2
    @. ws.tp = ws.wp + h/2 * ws.bp # Y3 = w + h/2 k2
    @. ws.tu = ws.wu + h/2 * ws.bu
    adv!(ws.cp, ws.tp, m, fh); adv!(ws.cu, ws.tu, m, fh) # k3
    @. ws.tp = ws.wp + h * ws.cp
    @. ws.tu = ws.wu + h * ws.cu
    E!(ws.tp, ws.tu) # Y4 = E(w + h k3)
    adv!(ws.ap, ws.tp, m, fh); adv!(ws.au, ws.tu, m, fh) # k4
    @. ws.tp = ws.wp + (ws.y2p - ws.wp)/3 + h/3 * (ws.bp + ws.cp)
    @. ws.tu = ws.wu + (ws.y2u - ws.wu)/3 + h/3 * (ws.bu + ws.cu)
    E!(ws.tp, ws.tu)
    @. p = ws.tp + h/6 * ws.ap
    @. u = ws.tu + h/6 * ws.au
end
```

- `adv!`: 5th order upwind flux form; all work arrays preallocated (`ws`); `etdrk4_step!` has the same structure.
- Programs: `fast_fb.jl` (module), `test_fast.jl` (accuracy of $\mathrm{e}^{h\mathcal{L}}, \varphi_k$), `run_experiments.jl` (data for the plots).

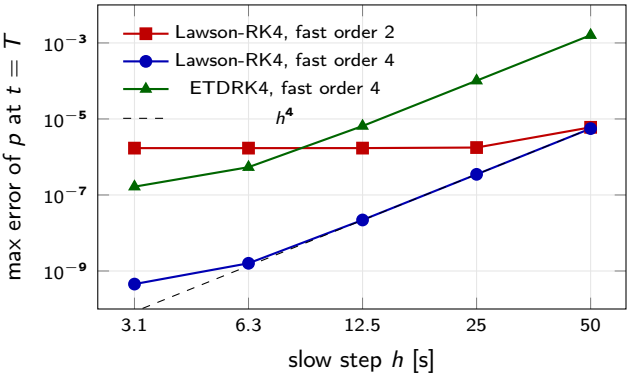
1D experiment: pressure pulse

$c = 350\,\mathrm{m/s}$, $U = 50\,\mathrm{m/s}$ ($M = 1/7$), $L = 1000\,\mathrm{km}$, 400 cells, $T = 10^4\,\mathrm{s}$: $(c+U)T = 4L$ and $(c-U)T = 3L$, so both acoustic pulses return to the start. Initial p: Gaussian, $u = 0$. Lawson-RK4, $h = 50\,\mathrm{s}$ ($ch/\Delta x = 7$, $Uh/\Delta x = 1$), fast integrator of order 4 with 8 substeps per $h/2$.



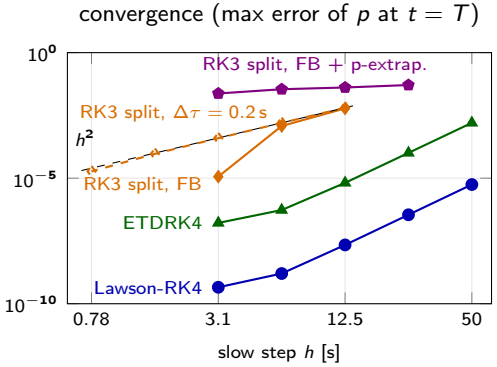
- $t = T/2$: the pulse has split into two half-amplitude pulses; the right-going one has traveled $2L$, the left-going one $1.5L$.
- $t = T$: back at the initial position; max. deviation $6 \cdot 10^{-6}$ (spatial dissipation of the upwind stencil plus time error).

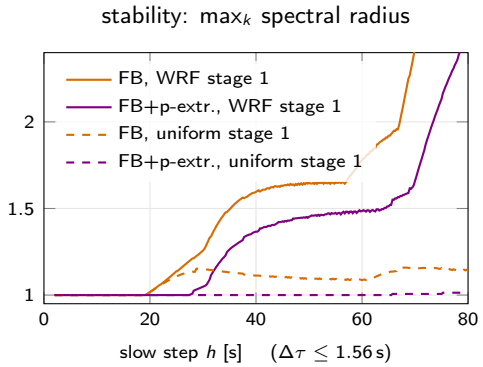
Convergence in the slow step: why the fast integrator needs high order



- Acoustic substep fixed by stability: $\tau = 1.56\,\mathrm{s}$, i.e. nsub $= 16, 8, 4, 2, 1$ per $h/2$.
- **Fast order 2:** error stalls at $1.7 \cdot 10^{-6}$ for $h \lesssim 25\,\mathrm{s}$: the 4th order slow scheme cannot be exploited.
- **Fast order 4:** fourth order in h from 50 down to 6 s ($1.6 \cdot 10^{-9}$); at $h = 3\,\mathrm{s}$ (nsub $= 1$) its fast error shows ($4.5 \cdot 10^{-10}$).
- ETDRK4: same order, much larger error constant here ($1.6 \cdot 10^{-3}$ vs. $5.6 \cdot 10^{-6}$ at $h = 50\,\mathrm{s}$); the curve flattens at $h \approx 3\,\mathrm{s}$ because of the fast error (nsub $= 1$).
- $\mathcal{L}, \mathcal{N}$ commute in this test: Lawson is then very accurate; ETDRK4 may profit when they do not.

Split-explicit RK3 (WRF) on the same test

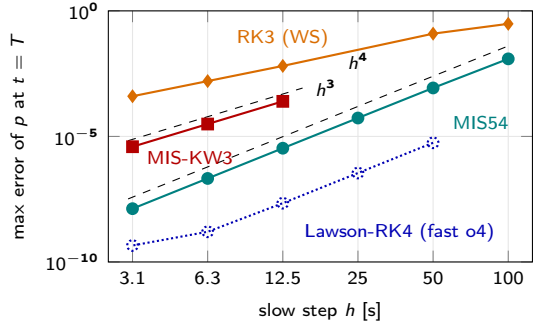




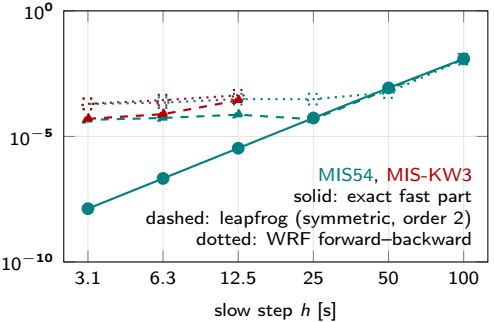
- Undamped FB: unstable for $h \geq 19\,\mathrm{s}$ ($ch/\Delta x \geq 2.7$) for both stage-1 variants. Pressure extrapolation ($\beta = 0.1$) stabilizes up to $h = 27.5\,\mathrm{s}$ with WRF's single stage-1 step ($c\,h/(3\Delta x) = 1.3$), up to $h = 66\,\mathrm{s}$ with $n_s/3$ small steps in stage 1.
- Second order in h when the fast part is resolved (dashed: factor 4 per halving), as predicted by the $\sin(\theta/2)/(\theta/2)$ term. The dip at $h = 3.1\,\mathrm{s}$ ($n_s = 2$) is a cancellation.
- Pressure extrapolation damps the *resolved* acoustic pulse: error $\propto \beta\Delta\tau$, it does not decrease with h.
- Equal cost at $h = 12.5\,\mathrm{s}$ (≈ 50 kick pairs per step): Lawson-RK4 (fast order 2) $2.6 \cdot 10^{-8}$, split-explicit RK3 $6.4 \cdot 10^{-3}$.

MIS methods on the 1D test

fast part (almost) exact



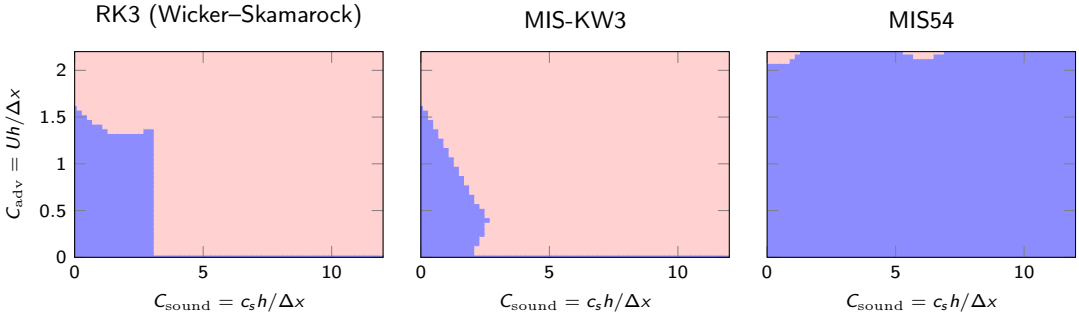
fast integrator in the MIS stages ($\tau \leq 1.56\,\mathrm{s}$)



- Orders 2, 3, 4 confirmed (MIS54: factor 16 per halving from $h = 100$ to $3.1\,\mathrm{s}$). MIS54 is stable for $h < 99\,\mathrm{s}$ ($ch/\Delta x < 13.9$) *without* divergence damping; RK3 is unstable from $h = 19.5\,\mathrm{s}$, MIS-KW3 is weakly unstable already at small h on this grid ($|\lambda|_{\max} - 1 = 2 \cdot 10^{-6}$ at $h = 6.25\,\mathrm{s}$, $2 \cdot 10^{-3}$ at $12.5\,\mathrm{s}$).
- The fast integrator must be symmetric and of high order: WRF's forward–backward step leaves an $\mathcal{O}(\tau)$ error that does not cancel between MIS stages (they start from combinations of earlier stages), leapfrog an $\mathcal{O}(\tau^2)$ error; they stall at $2 \cdot 10^{-4}$ and $5 \cdot 10^{-5}$.
- MIS54 has the fast work of RK3 ($\sum d_i = 1.88$) but 5 evaluations of $\mathcal{N}$; in this test (commuting $\mathcal{L}$, $\mathcal{N}$) Lawson-RK4 is still more accurate.

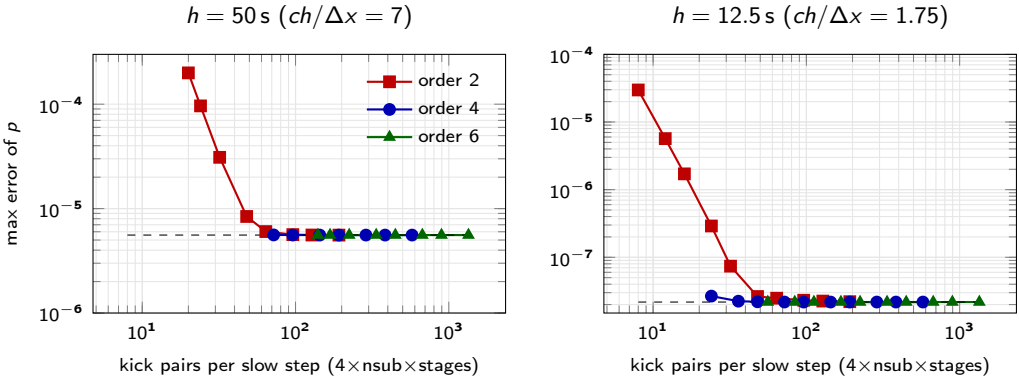
Example from Knoth–Wensch: stability regions for linear acoustics

$u_t + Uu_x = -c_s\Pi_x$, $\Pi_t + U\Pi_x = -c_s u_x$ (Knoth–Wensch 2014, eq. (10)); C-grid, central differences for the acoustic terms, 3rd order upwind advection; exact integration of the fast stages; $\max_k |\lambda(S(C_{\mathrm{adv}}, C_{\mathrm{sound}}, k))|$ over 40 Fourier modes.



- Blue: stable. Pure advection ($C_{\mathrm{sound}} = 0$): $C_{\mathrm{adv}} \leq 1.63$ for the 3-stage methods (RK3 with 3rd order upwinding), 2.06 for MIS54.
- Without damping RK3 and MIS-KW3 become unstable for $C_{\mathrm{sound}} \gtrsim 3$ and $\gtrsim 2$ (exact fast part); MIS54 is stable on the whole range (checked with 400 modes), it was selected for this test.
- Same picture on our grid (4th order acoustics, 5th order upwind): MIS54 stable up to $ch/\Delta x = 13.9$, RK3 only up to 2.7.

Work–precision of the fast integrator (Lawson-RK4)

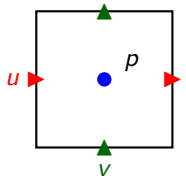


Dashed: error of Lawson-RK4 with exact $\mathrm{e}^{h\mathcal{L}/2}$ (slow-scheme error, the best possible).

- The fast integrator is accurate enough once its error is below the dashed line; missing points: substeps too large (unstable).
- The gain of a high-order fast integrator grows as the slow step (and tolerance) get smaller; order 6 does not pay off against the 4th order slow scheme.

Two dimensions

$$\partial_t \begin{pmatrix} p \\ u \\ v \end{pmatrix} = - \begin{pmatrix} \rho_0 c^2 (\partial_x u + \partial_y v) \\ \frac{1}{\rho_0} \partial_x p \\ \frac{1}{\rho_0} \partial_y p \end{pmatrix} - (U \partial_x + V \partial_y) \begin{pmatrix} p \\ u \\ v \end{pmatrix}.$$



- Arakawa C-grid: p at centers, u on vertical faces, v on horizontal faces.
- Advection of u and v requires *different* stencils in x and y; high-order upwind per direction.
- Fast operator $\mathcal{L}$: 2D wave operator; $\mathrm{e}^{h\mathcal{L}}$, φ_k again by the substepping integrator (the FFT is only used for reference solutions).
- Tests: acoustic pulse in a uniform wind (U, V); rotated wind to test directional splitting errors.

Expected observations (hypotheses to check)

- Split-explicit (measured in 1D): the frozen slow forcing limits RK3 (WRF) to second order in h; without divergence damping it is unstable for $ch/\Delta x \gtrsim 2.7$, pressure extrapolation stabilizes it but damps resolved acoustic signals.
- ETD/Lawson: acoustic stability for arbitrary h, order determined by the outer RK/ETD scheme.
- Resonance steps $ckh \approx 2\pi n$: visible spikes in the error for Lawson/ETD; reduced for averaging; removed for resonance-based schemes.
- When $[\mathcal{L}, \mathcal{N}] \neq 0$ (variable wind, flux-form advection), the commutator gives the leading splitting error; averaging/resonance corrections remove it to higher order.
- In 2D: extra error from the directional splitting of $\mathcal{N}$; cost advantage of FFT-based exponentials shrinks with non-periodic boundaries.

(first item measured; the others to be checked with variable wind and in 2D)

Summary and outlook

- The linear acoustic–advection system isolates the main stiffness/multirate difficulty of NWP dynamical cores.
- **Split-explicit** methods are ETD1-type methods with substepped approximations of $\mathrm{e}^{h\mathcal{L}}$ and $\varphi_1(h\mathcal{L})$.
- **Lawson**, **averaging** and **resonance-based** methods treat the fast flow by an exponential and differ in how the slow interaction integral is approximated.
- One **high-order forward–backward (leapfrog/composition) integrator** delivers $\mathrm{e}^{h\mathcal{L}}$ and all φ_k through polynomial sources; its order must match the slow scheme because τ is stability limited.
- Application (staggered grid, 5th order upwind advection):
  - Lawson-RK4 and ETDRK4 with a fast integrator of order 4: order 4 in h up to $ch/\Delta x = 7$; with order 2 the error stalls.
  - Split-explicit RK3 (WRF): second order, divergence damping needed beyond $ch/\Delta x \approx 2.7$, orders of magnitude less accurate at equal cost.
  - MIS54: order 4, stable without damping up to $ch/\Delta x \approx 14$, needs a symmetric high-order fast integrator in its stages.
- Outlook: variable coefficients, orography/terrain-following coordinates, nonlinear (Euler) equations, IMEX/semi-implicit comparison, parallel efficiency; variable wind and 2D ($[\mathcal{L}, \mathcal{N}] \neq 0$).

Thank you for your attention!

References (to be completed) I

[1] J. B. Klemp, R. B. Wilhelmson, *The simulation of three-dimensional convective storm dynamics*, J. Atmos. Sci. 35 (1978).

[2] L. J. Wicker, W. C. Skamarock, *Time-splitting methods for elastic models using forward time schemes*, Mon. Wea. Rev. 130 (2002).

[3] J. B. Klemp, W. C. Skamarock, J. Dudhia, *Conservative split-explicit time integration methods for the compressible nonhydrostatic equations*, Mon. Wea. Rev. 135 (2007).

[4] M. Baldauf, *Linear stability analysis of Runge–Kutta-based partial time-splitting schemes for the Euler equations*, Mon. Wea. Rev. 138 (2010).

[5] O. Knoth, J. Wensch, *Generalized split-explicit Runge–Kutta methods for the compressible Euler equations*, Mon. Wea. Rev. 142 (2014).

[6] T. P. Bauer, O. Knoth, *Extended multirate infinitesimal step methods: Derivation of order conditions*, J. Comput. Appl. Math. 387 (2021) 112541.

[7] O. Knoth, R. Wolke, *Implicit-explicit Runge–Kutta methods for computing atmospheric reactive flows*, Appl. Numer. Math. 28 (1998).

References (to be completed) II

[8] J. M. Sexton, D. R. Reynolds, *Relaxed multirate infinitesimal step methods for initial-value problems*, arXiv:1808.03718 (2019).

[9] M. Hochbruck, A. Ostermann, *Exponential integrators*, Acta Numerica 19 (2010).

[10] S. M. Cox, P. C. Matthews, *Exponential time differencing for stiff systems*, J. Comput. Phys. 176 (2002).

[11] J. D. Lawson, *Generalized Runge–Kutta processes for stable systems with large Lipschitz constants*, SIAM J. Numer. Anal. 4 (1967).

[12] J. Wensch, O. Knoth, A. Galant, *Multirate infinitesimal step methods for atmospheric flow simulation*, BIT 49 (2009).

[13] S. Gaudreault, J. Pudykiewicz, *An efficient exponential time integration method for the numerical solution of the shallow water equations on the sphere*, J. Comput. Phys. 322 (2016).

Please check the bibliographic details and add the specific references for averaging and resonance-based methods you intend to follow.